# Supertrend

1h trigger | Freqtrade source review | long-only

## The idea

Imported Freqtrade strategy running on 1h. Its indicator layer uses the source does not expose recognised TA calls. The backtest snapshot labels it 'promising'.

## How the strategy makes a decision

1. `populate_entry_trend` prepares the entry flag from the strategy dataframe.
2. Source entry gates: volume > 0; dataframe[f'supertrend_3_buy_{buy_m3.value}_{buy_p3.value}'] == 'up'; dataframe[f'supertrend_1_buy_{buy_m1.value}_{buy_p1.value}'] == 'up'; dataframe[f'supertrend_2_buy_{buy_m2.value}_{buy_p2.value}'] == 'up'.
3. Warm-up requirement: 199 candles. Signals are long-side only in this source.

## How a profit could arise

The edge exists only if this exact combination of indicator values continues to select favourable price movement after fees, gaps and the strategy's risk rules. The source describes a signal, not proof of an edge.

## How the trade ends

source stoploss -0.265; minimal_roi {'0': 0.087, '372': 0.058, '861': 0.029, '2221': 0}; trailing stop enabled.

## A practical signal trace

After a closed 1h candle, Freqtrade computes the source dataframe and `populate_entry_trend` may set an entry flag when the listed fields satisfy its boolean predicate. The backtest snapshot then applies its configured risk layer, which may differ from these source defaults.

## How Freqtrade backtests it

### Historical replay

Freqtrade replays stored OHLCV candles in time order. Strategies calculate a dataframe with `populate_indicators`, then set entry and exit columns. A signal from a completed candle is evaluated by the engine's candle-based fill model; it cannot reconstruct tick order, queue position or historic order-book depth.

## What this source contributes

1. Indicator method: `populate_indicators`; recognised calls: none detected.
2. Entry method: `populate_entry_trend`; observed input fields: `volume`.
3. Exit method: `populate_exit_trend`; observed input fields: `volume`.

## How the backtest snapshot was priced

REPORT-ALL uses KuCoin ETH/USDT spot, 19 Aug 2023 to 18 Aug 2026, 5,000 USDT starting balance and a 0.1% fee per side. It explicitly says its configuration overrode every strategy's `stoploss` with -10% and `minimal_roi` with 30%; reported exits therefore do not validate the author's source risk settings.

## Causality and intrabar limits

Freqtrade lookahead-analysis was clean over 20 signals. Lookahead testing still does not remove OHLC ambiguity for ROI, trailing stops and intrabar stop/target collisions. Re-run material candidates with `--timeframe-detail 1m` before treating a candle-level result as executable.

## What makes the result credible?

A source review and a transparent replay narrow the gap to live trading; neither establishes profitability. Read the evidence and its boundaries together.

## What the backtest snapshot can substantiate

| TIMEFRAME | TRADES | P/L (USDT) | PF | SHARPE | MAX DD |
|---|---|---|---|---|---|
| 1h | 124 | +422 | 1.087 | 0.07 | 19.25% |

Source run: `all-20260819-095325`. Values above are copied from the provided backtest snapshot.

## Built-in safeguards and their limits

Report classification: promising. Risk style inferred from source: structural-swing. Freqtrade lookahead-analysis was clean over 20 signals. The existing report is one historical run, not a frozen-parameter chronological holdout or a live-fill study.

## The key live-trading gap

Live prices, spreads, partial fills, exchange rejection and outages are outside the historical candle model. In particular, the report did not reproduce historical KuCoin order-book depth.

## Evidence needed before live use

1. Confirm continuous candles, exact exchange fee tier, pair rules and source/config risk settings; then remove the report override and rerun.
2. Freeze parameters before a chronological out-of-sample period. Review trade count, net P/L after costs, drawdown and sensitivity to higher fees.
3. Paper trade the frozen strategy, compare signal timestamps with fills, and explain material drift before risking capital.